

National University of Computer and Emerging Sciences, Lahore Campus



Course: Signals & Systems
Program: BS(Electrical Engineering)
Duration: 60 Minutes
Paper Date: 22nd February, 2017
Section: ALL
Exam: Midterm-I

Course Code: EE206 223
Semester: Spring 2017
Total Marks: 40
Weight: 15%
Page(s): 1
Roll No:

Instruction/Notes: Carefully read the paper and attempt each question (only once). Attempt all parts of a question together.

Question #1: CLO (1)

10 Marks

Mathematical expression of a continuous time signal is given below:

$$x(t) = 2[u(t+6) - u(t+3)] + (-t-1)[u(t+3) - u(t+1)]$$

- Sketch $x(t)$
- Sketch $\frac{1}{2}x(-3t-6)$

Question #2: CLO (1)

10 Marks

Classify the system below as:

- Linear or non-linear
- Time variant or time invariant

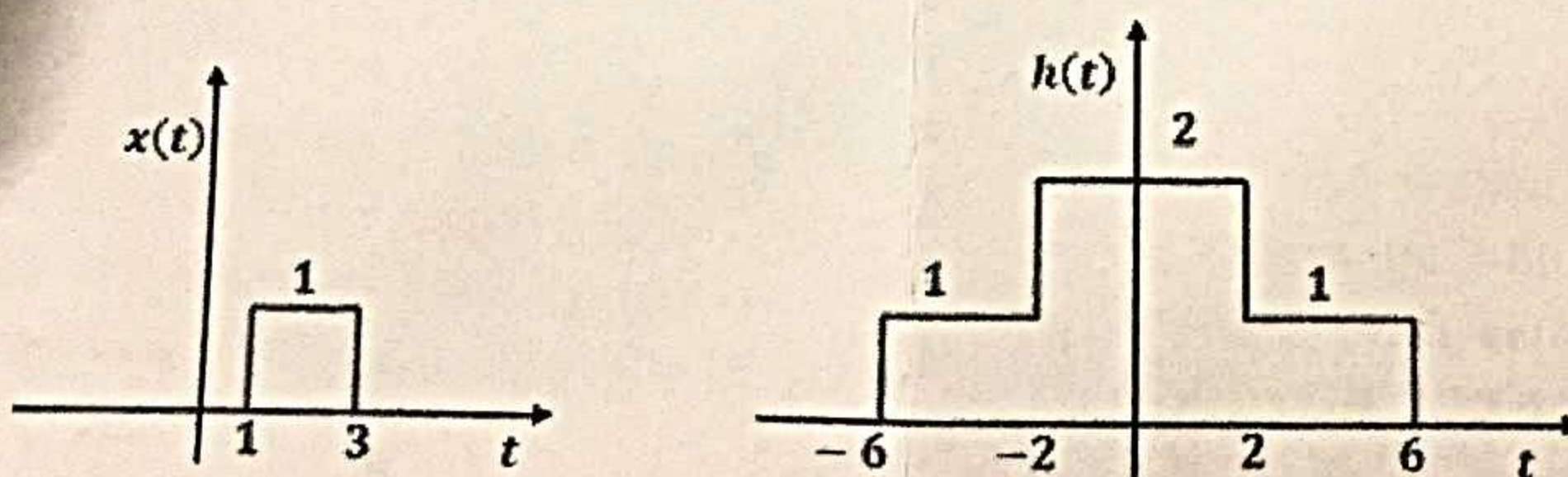
$$y(t) = \cos(x(t))u(t)$$

Provide a concise mathematical justification for your answer.

Question #3 CLO (2)

20 Marks

Two continuous time signals $x(t)$ and $h(t)$ are given below.



Use graphical techniques to find and sketch $y(t) = x(t) * h(t)$.

at $-b$
 $t-b$

at

$3t+5$

$t+5$

$x(t+3)$

$t+5$